Azhar Amir Kimanje

XR & AI Engineer · Curriculum Vitae

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Simplicity is Sophistication. I'm an XR and AI engineer with a unique perspective on improving multi-sensory cognitive decision-making in medical training and challenge-based education.

EDUCATION

B.A., Computer Science and Engineering · Seattle Pacific University, Seattle, WA — Sep 2020 – Jun 2024

EXPERIENCE

Lead Interoperability Engineer · Exonicus Inc., Seattle, WA — Jun 2024 – Present

Real-time XR trauma simulation that trains combat medics to save lives. Also Software Engineer, XR.

- Integrated physiology engines, virtual patients and military training standards (HLA, JETS), following a Modular Open Systems Approach (MOSA), so separate simulation systems share one patient in real time.
- Engineered a custom API plugin for real-time medical data communication, getting training to military medics **11 months** sooner.
- Shipped MilExTS (Meta Quest Store) and Exonicus Scenario Builder (Microsoft Store) to put trauma training in medics' and educators' hands.
- Optimized data handling with Unity DOTS to process **thousands of medical injury combinations** with minimal cache misses, and standardized naming across large medical datasets so teams could use them.
- Led IEEE quality assurance for mixed reality game-engine plugins and designed CI/CD pipelines with **a team of 5**.
- Drove collaboration in a **five-company** Simulation Interoperability Working Group cohort, supporting LVC training commercialized in **six countries**.
- Partnered with US Army DEVCOM on XR usability studies, and taught educators to build no-code AR trauma scenarios at IMSH 2026.

Director of Mixed Reality and AI Outreach · AVELA Non-profit, Seattle, WA — Jun 2024 – Present

Director of AI × XR Labs, bringing computing and XR education to underrepresented and rural students.

- Led on-device AI research with Apple MLX so student data stays on the device, shipping the result as CourseSLM on the App Store.
- Secured and executed a **$200K Apple Community Education Initiative award** (2026), an **$80K Apple grant** (2025) and a **$75K UW Student Technology Fee grant** (2024) to scale XR and AI education.
- Orchestrated XR programs in data activism and environmental awareness that reach **3,200+ STEM students a year**, with partners including Apple Education and SPEEA ACE.
- Designed the acquisition pipeline for **30 Quest 3 headsets**, **60 laptops** and **30 iPads** so every student had hands-on access.
- Published a teaching kit on agentic AI coding on Apple's Education Community to help developers and educators adopt AI responsibly.

Founder & CEO · Sablesync, Seattle, WA — Nov 2023 – Present

Spatial computing that anyone can use; ML fine-tuning and digital-twin R&D.

- Launched **four Apple apps** that make 3D capture accessible without specialist gear: Sablesync RC, Omni, 3D Reconstruction and Q-AR (see Shipped Apps).
- Prototyped an emergency-evacuation digital twin with Unity NavMesh AI, automating floor plan to 3D model with NVIDIA Omniverse, from R&D to MVP in **one month** with **3 engineers**.
- Led **3 engineers** on real-time AR hardware communication, cutting system integration time **50%** and development cycles **40%**.
- Automated testing for AR hardware components and streamlined deployment with Oculus Integration, reducing firmware errors.

Technical Mixed Reality Lead Instructor · SPEEA Aerospace Career Enhancement (ACE), Tukwila, WA — Dec 2024 – Present

- Teach XR with computer-vision models for Gaussian splats, preparing aerospace professionals for spatial computing careers.
- Designed drone-based solutions for spatial planning and agriculture, and lead VR and AR initiatives for aerospace and sustainability projects.
- Teach Mixed Reality Applications: VR for Aerial Search & Rescue, an instructor-led course preparing crews for aerial search and rescue.

Software Engineer, XR – Design Intelligence · IA Interior Architects, Seattle, WA — Jun 2024 – Sep 2024

- Created **10+ XR space plans** in Unity and C#, and verified all built-environment XR models in Unity and Revit, so architects could test spaces before building.
- Researched generative AI solutions in Python to speed up cost estimation, cutting project time **20%**.

Augmented Reality Developer, Extern · Snap Inc., Seattle, WA — Mar 2024 – Apr 2024

- Designed Lens Studio 5.5 AR experiences for sports businesses, and wrote product requests and system design documents to align customers and stakeholders.

Lead Feature Engineer, Extended Reality Course Designer · AVELA Non-profit, Seattle, WA — Nov 2023 – Jun 2024

- Built AVELA's first **40-hour** XR curriculum to make computing approachable, raising student STEM engagement **25%**.
- Trained conversational AI agents in XR as teaching assistants, and delivered **80 hours** of ML, AI and data-activism curriculum.

STEM Lead Instructor · AVELA Non-profit, Seattle, WA — Sep 2023 – Nov 2023

- Wrote the grants that brought XR curriculum and hardware into AVELA, and taught Python, data activism and computing for environmental awareness.

Product Owner, Software Engineering Capstone · Seattle Pacific University — Sep 2023 – Jun 2024

- Led SPU GradPro, a 3D simulation of the CS degree on WebGL, adding the university's first conversational AI scene with generative-AI NPCs of CS professors; won **1st** Place, Top Poster Presentation.

Product Management Intern · YouthForce, Boys & Girls Clubs of America — Nov 2023 – Jan 2024

- Ran **5 beta tests** of the Youthforce XR Lounge with **30 testers** at a time, turning feedback into features for the final release on Meta App Labs.

Software Engineering Intern · YouthForce, Boys & Girls Clubs of America — Jul 2023 – Dec 2023

- Led full-stack development of the first XR lounge for the National Boys & Girls Club after winning the Youthforce Coding Challenge, shipping the MVP **60%** faster and doubling rendering to **120 fps**.

Freshman Counseling Team Lead, then Intern · SPU Undergraduate Admissions — Dec 2020 – Jun 2024

- Led **12 interns** making **1,200+ outbound calls a week** to help students choose SPU, contributing to a **35%** enrollment increase and a **25%** rise in minority admissions to the School of Engineering.

Early Insight Diversity Series · Citi — Mar 2024 – Apr 2024

RESEARCH & DEVELOPMENT

TCCC SBIR Phase II and IIE · Exonicus Inc. · Lead Interoperability Researcher — Jun 2024 – May 2026
Principal Investigator: Dr. Kyle Couperus

- Led research on a real-time XR medical simulation system, improving data interoperability, latency and reliability so military training outcomes reflect the field.

CourseLLM · AVELA Non-profit · Principal Investigator — Jun 2025 – Sep 2025

- Led **3 PhD candidates and 6 interns** fine-tuning hardware-constrained language models with Apple MLX for data sovereignty, delivering a data-activism lesson plan with Apple's Community Education Initiative and shipping CourseSLM.

Gaussian Splats for Storytelling · AVELA Non-profit · Principal Investigator — Jan 2025 – Aug 2025

- Architected a deep-learning workflow that turns 360° drone video into true-to-depth XR environments, published openly with Bellevue XR Labs under the Washington Open Common Education License.

Avatar Controller Toolkit (XR ACT) · ISIA Lab, Université de Mons · Research Contributor — Jul 2024 – Present
Principal Investigator: Dr. Kevin El Haddad

- Advancing real-time facial landmark detection so XR avatars express emotion more accurately and efficiently.

BioDepot Workflow Builder · Morphic Lab, University of Washington Tacoma · Research Contributor — Sep 2024 – Present
Principal Investigator: Dr. Ka Yee Yeung

- Contributing to a GUI that auto-installs workflow components so non-programmers can reproduce gene research, with Jupyter and Cytoscape support.

Wordplay · LCI Lab, University of Washington · Research Contributor — Sep 2023 – Present
Principal Investigator: Dr. Amy J. Ko

- Contributing to an accessible, multilingual programming platform that makes computing education more equitable.

PUBLICATIONS

- **Rapid Military Medical Experimentation Using a Configurable Mixed Reality Scenario Platform.** Co-author. Presented at the Military Health System Research Symposium (MHSRS) 2026.
- **Hybrid Physical-Digital Training Environments for Combat Casualty Care.** First author. Submitted to *Military Medicine*, 2026.
- **Build, Train, Track: A Proposed Extended Reality Framework for Military Medical Experimentation.** Co-author. Submitted to *Military Medicine*, 2026.
- **Welcome to the Agentic Era: Strategic AI Implementation for Developers.** First author. Teaching kit, Apple Education Community, Community Education Initiative, Oct 2026. CC0 license.
- **From Toy to Tool: An Investigation on Student AI Use Cases and Effects.** Co-author. Presented at the World Engineering Education Forum & Global Engineering Deans Council (WEEF & GEDC) 2026, Cartagena, Colombia.
- **Tool or Toy? Student Views on ChatGPT in Culturally Responsive Computing Education: A Preliminary Study.** IEEE Xplore, 2025.
- **Gaussian Splats for Storytelling.** First author. OER Commons, with Bellevue College XR Lab, 2024.
- **Lessons Learned from Creating a Standard for Medical Simulation Interoperability.** Co-author. SISO SIW 2025, Paper #15.

SHIPPED APPS

CourseSLM · On-device AI for STEM learning · AVELA	Apple App Store
MilExTS · XR trauma-care training for military teams · Exonicus	Meta Quest Store
Exonicus Scenario Builder · Trauma scenario authoring · Exonicus	Microsoft Store
Sablesync RC · LiDAR room capture to blueprints and 3D models · iPhone, iPad, Mac, Vision Pro	Apple App Store
Sablesync Omni · Photos to USDZ 3D models · iPhone, iPad, Vision Pro	Apple App Store
Sablesync 3D Reconstruction · On-device 3D reconstruction from images · Mac	Mac App Store
Sablesync Q-AR · USDZ model and animation viewer · iPhone, iPad, Mac	Apple App Store

GRANTS LED AND EXECUTED

Exonicus Inc. · XR trauma simulation and interoperability R&D · May 2024	**$1,900,000**
Exonicus Inc. · XR trauma simulation and interoperability R&D · Sep 2025	**$750,000**
Apple Community Education Initiative · On-device AI and AI education outreach across Washington State · Summer 2026	**$200,000**
Apple · On-device AI development for the AVELA community · Summer 2025	**$80,000**
UW Student Technology Fee (STF) · Expanding XR education across UW and Seattle · Sep 2024	**$75,000**
UW Student Technology Fee (STF) · XR instruction and hardware · Feb 2024	**$65,000**
Total	**$3,070,000**

AWARDS

1st Place, University of Washington Tech Showcase · Jun 2024	**$1,000**
1st Place, Top Poster Presentation Award · Seattle Pacific University · Jun 2024	**Winner**
3rd of 200, Global Eduhacks · University of Washington · Prebrief · Jun 2024	**$500**
1st, Snap Inc. Master of Augmented Reality · Apr 2024	**Internship interview**
6th of 150, Amazon AWS & Inrix Hackathon · Wanderus · Dec 2023	**Internship interview**
1st Place, Youthforce Coding Challenge · National Boys & Girls Club · Jul 2023	**$700**

SCHOLARSHIPS

Director's Grant · Seattle Pacific University · 2020 – 2024	**$8,000 / year**
Washington Bridge Grant · Seattle Pacific University · 2020 – 2024	**$3,000 / year**
Culture Tech Legacy Google, Masterclass and LinkedIn Learning Scholarships · 2024	**Fully paid certifications**

MEDIA COVERAGE

- **AVELA researchers present at WEEF & GEDC 2026**, AVELA UW, Oct 2026. Co-author of the presented paper.
- **Welcome to the Agentic Era**, Apple Education Community, Oct 2026. Author of the teaching kit.
- **Startup Spotlight Sessions, Seattle Startup Hall**, Exonicus, Aug 2026. Represented Exonicus as Lead Engineer.

- **Exonicus at the IMSH 2026 Vendor Showcase**, HealthySimulation.com video, Jan 2026. On-camera product walkthrough.
- **IMSH 2026, San Antonio**, Jan 2026. Workshop instructor ("From Zero to Scenario") and on-site team at Exonicus Booth #104.
- **MilExTS at the AUSA Annual Meeting**, Exonicus, Oct 2025. Led live demonstrations and discussions in the Army Sustainment Area.
- **Latvia builds opportunities for cooperation with the U.S. military**, LTV1 Panorāma (Latvian national television), Oct 15, 2025. Featured on air demonstrating MilExTS at AUSA 2025.
- **Gaussian Splat for Storytelling**, XR Lab @ Bellevue College, YouTube, Sep 2025. Featured describing my drone-based Gaussian splat approach.
- **FIRST Responder SBIR Phase II Touchpoint**, Fort Gordon, Georgia, Jul 2025. Demonstrated the Exonicus Trauma Simulator.
- **Meet Student Entrepreneur Azhar Kimanje**, SPU Stories, Jul 2024. Feature article.
- **Meet the Class of 2024**, Seattle Pacific University Commencement, Jun 2024. Profile.
- **Live@5 Guest List**, SynED. Guest.

LEADERSHIP & VOLUNTEER WORK

- Regional Executive Board Finance Chair, National Society of Black Engineers, Puget Sound region, Jul 2024 – Nov 2024. Managed corporate partnerships and conference financial planning (**$160,000**).
- Founder & President, NSBE Seattle Pacific University Chapter, Sep 2023 – Jun 2024.
- Program budget, Director of Mixed Reality Outreach, AVELA Non-profit, Jun 2024 – Present (**$150,000**).
- **Volunteer:** XR Technical Support and Student Volunteer, National Boys & Girls Club (current) · Scholarship Essay Support and Coding Instructor, AVELA Non-profit (current) · Welcome Team, Seattle Pacific University (2023 – 2024).

ORGANIZATIONS

- Simulation Interoperability Standards Organization (SISO), Medical Modeling & Simulation Community, through Exonicus, 2025 – Present.
- Alliance for OpenUSD (AOUSD), through Sablesync, 2025 – Present.
- Apple Community Education Initiative (CEI), AI education partner, through AVELA.
- Northwest XR EDU Network (NW XR EDU), through AVELA, 2024 – Present.

CERTIFICATES

Microsoft	Azure Digital Twins; Connect IoT Data to Mixed Reality with Azure Digital Twins and Unity; Introduction to Azure Digital Twins; Mixed Reality and Azure Digital Twins; Office Specialist: Excel Associate (2024)
Unity	VR Development; Shaders and Materials; Data-Oriented Design and Entity Component Modeling; Junior Programmer (2023 – 2024)
LinkedIn Learning	CISSP: Identity and Access Management; Generative AI for Product Managers; Web Programming Foundations; C# Development (2023 – 2024)
Google · Meta · Snap · Exonicus · Other	Connect and Protect: Networks and Network Security; Foundations of Cybersecurity; Play It Safe: Manage Security Risks (2024); Meta Spark Foundations of AR (2024); Snap Master of AR Development (2024); OPSEC Awareness for Military Members and DoD Employees (2024); 3D Graphics on the Web with Three.js

SKILLS

Programming	C# (5+ years), C++, Swift, Python, C, SQL, Java
Spatial & 3D	Unity (DOTS, NavMesh), Unreal, OpenXR, RealityKit, ARKit, ARCore, USDZ / OpenUSD, LiDAR, Gaussian splatting
AI & vision	MLX fine-tuning, small language models, computer vision, deep learning, NLP, rapid prototyping
Simulation	HLA, JETS, Modular Open Systems Approach (MOSA), real-time data synchronization, CI/CD
Spoken languages	Luganda (native), English (native), Spanish (elementary)